(b) Equivalent mass of a oxidising agent $= \dfrac{\text{molar mass}}{\text{number of moles of electrons gained by one mole of the reducing agent}}$

$= \dfrac{294.2 \text{ g mol}^{-1}}{6 \text{ eq mol}^{-1}}$

$= \mathbf{49.0 \text{ g eq}^{-1}}$

5.

Element	**Percentage Composition**	**Atomic mass**	**Relative no. of atoms = $\frac{\text{Percentage}}{\text{Atomic mass}}$**	**Simple ratio**
C	54.55 %	12	54.55/12 = 4.55	4.55 / 2.27 = 2
H	9.09 %	1	9.09 / 1 = 9.09	9.09 / 2.27 = 4
O	36.36 %	16	36.36/16 = 2.27	2.27/2.27 = 1
Empirical formula (C_2H_4O)				

6.

Element	**Percentage Composition**	**Relative no. of atoms = $\frac{\text{Percentage}}{\text{Atomic mass}}$**	**Atomic mass= $\frac{\text{Percentage}}{\text{Relative no. of atoms}}$**	**Simple ratio**
X	32 %	2	16	4
Y	24 %	1	24	2
Z	44 %	0.5	88	1
Empirical formula ($X_4 Y_2 Z$)				

Calculated empirical formula mass $= (16 \times 4) + (24 \times 2) + 88$

$= 64 + 48 + 88 = 200$

$n = \dfrac{\text{molar mass}}{\text{calculated empirical formula mass}}$

$\therefore n = \dfrac{400}{200}$

$= 2$

$\therefore$ **Molecular formula $(X_4Y_2Z)_2$** $= \mathbf{X_8Y_4Z_2}$

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